

SIMATS ENGINEERING

ASSESSMENT TOOL 2

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COURSE NAME: Cybersecurity Incident Handling And Response
COURSE CODE: CSA60

CO5: *Implement database operations using query processing, optimization, and transaction management to ensure consistency and reliability. (BL3, BL4)*

Activity Tool : Pseudocode Writing Task (Algorithm Design) (Medium)
Assessment Tool Weightage: 20%

Code of Conduct:

I _____shaikreehan.192565071_____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Shaik Reehan

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Incident Classification and Severity Assessment	25%	Accurately classifies incidents and determines severity based on impact, affected assets, data sensitivity, urgency, and business criticality with clear justification.	Correctly classifies most incidents and determines appropriate severity with minor gaps.	Provides basic classification and severity assessment but misses important factors.	Unable to correctly classify incidents or determine appropriate severity.
Incident Ticket Creation and Documentation	20%	Creates complete and accurate tickets containing incident details, timestamps, affected assets, IOCs, evidence, assigned personnel, actions, and status updates.	Creates well-structured tickets with most required information and minor omissions.	Creates basic tickets but important information or evidence is missing.	Unable to create accurate or complete incident tickets.
Incident Prioritization, Tracking, and Escalation	25%	Effectively prioritizes, tracks, updates, and escalates incidents based on severity, SLA requirements, business impact, and escalation procedures.	Correctly prioritizes and tracks incidents with minor gaps in escalation or SLA handling.	Performs basic tracking and prioritization but misses important escalation or SLA requirements.	Unable to appropriately prioritize, track, or escalate incidents.
SOC Coordination and Incident Resolution	15%	Effectively coordinates with SOC and relevant technical teams, documents response actions, verifies resolution, and properly closes incident tickets.	Coordinates with relevant teams and completes incident resolution with minor omissions.	Demonstrates limited coordination and incomplete resolution or closure procedures.	Unable to coordinate effectively or complete the incident resolution process.
Reporting, Documentation, and Presentation	15%	Provides clear, professional incident reports and ticket records with complete evidence, activity history, SLA status, resolution details, and conclusions.	Documentation is well organized and includes most required information with minor issues.	Documentation is partially organized and lacks important details or evidence.	Documentation is incomplete, poorly organized, and lacks clear incident records or conclusions.

1. Multiple Incident Tracking and Resource Allocation

Introduction

A Security Operations Center (SOC) may receive multiple security incidents simultaneously, each with different levels of severity and business impact. The incident response manager must ensure that every incident is properly classified, documented, prioritized, assigned, monitored, and resolved.

A centralized **ticket management system** should be used as the primary source of truth for tracking all incidents. It provides visibility into incident status, ownership, SLA deadlines, escalation requirements, evidence, and response activities.

A. Incident Classification

The first step is to classify each incident according to its nature and security impact.

Common incident categories include:

- Phishing
- Malware
- Unauthorized access
- Credential compromise
- Data breach
- Denial-of-Service attack
- Vulnerability exploitation
- Insider threat
- Database compromise

Each incident should also be assessed using:

1. **Business impact**
 2. **Affected assets**
 3. **Data sensitivity**
 4. **Number of affected users**
 5. **Urgency**
 6. **Likelihood of further damage**
 7. **Business criticality of the affected system**
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B. Severity Assessment

A four-level severity model can be used:

Severity	Description	Example
P1 – Critical	Severe or ongoing business/security impact	Active database compromise
P2 – High	Significant security risk requiring rapid response	Compromised administrator account
P3 – Medium	Limited impact requiring timely investigation	Malware on one workstation
P4 – Low	Minimal impact and low urgency	Blocked phishing attempt

Severity should not be based only on the type of attack. The actual impact and criticality of the affected asset must also be considered.

C. Classification of the Five Simultaneous Incidents

Suppose the SOC receives these five incidents:

Ticket	Incident	Severity	Priority	Reason
INC-001	Critical database compromise	P1 – Critical	Highest	Sensitive data and critical business system affected
INC-002	Administrator credential compromise	P2 – High	Very High	Could provide privileged access
INC-003	Malware on employee workstation	P2/P3	High/Medium	Potential endpoint compromise and lateral movement
INC-004	Successful phishing attempt	P3 – Medium	Medium	User credentials may be at risk
INC-005	Blocked phishing attempt	P4 – Low	Low	Attack was prevented with no confirmed compromise

The **database compromise receives immediate attention** because it has the highest potential business and security impact.

D. Incident Ticket Creation

A separate ticket should be created for every incident.

Each ticket should contain:

Basic Information

- Incident ID
- Incident category

- Date and time detected
- Date and time reported
- Reporter
- Affected user/system
- Severity
- Priority
- Current status

Technical Information

- Source and destination IP addresses
- Hostnames
- User accounts
- Indicators of Compromise (IOCs)
- Malware hashes where applicable
- Suspicious URLs/domains
- Relevant log entries
- Screenshots or other evidence
- Vulnerabilities involved

Response Information

- Assigned analyst
- Assigned team
- Investigation actions
- Containment actions
- Eradication actions
- Recovery actions
- Escalation history
- Communication history
- SLA deadline
- Status updates

This creates a complete audit trail of the incident.

E. Ticket Status Tracking

A standard workflow should be implemented:

New → Triage → Assigned → Investigating → Containment → Eradication → Recovery → Monitoring → Resolved → Closed

The ticket management system should automatically record timestamps whenever an incident changes status.

For example:

10:00 – Incident detected

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10:05 – Ticket created

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10:10 – Analyst assigned

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10:20 – Investigation started

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10:45 – System isolated

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12:00 – Threat removed

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13:00 – System restored

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14:00 – Incident resolved

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Next day – Ticket closed after verification

This history allows the SOC manager to calculate response and resolution times.

F. Resource Allocation

Resources should be allocated according to **risk and business impact**, rather than simply handling incidents in the order they arrive.

P1 – Critical Database Compromise

Assign:

- Senior incident responder
- SOC analyst

- Database administrator
- Network/security engineer
- Threat-hunting specialist
- Incident response manager
- Management/business owner

Actions:

1. Immediately investigate the compromise.
2. Isolate affected database systems if required.
3. Preserve forensic evidence.
4. Identify compromised accounts.
5. Determine whether data was accessed or exfiltrated.
6. Reset or revoke compromised credentials.
7. Remove attacker persistence.
8. Patch exploited vulnerabilities.
9. Monitor for additional malicious activity.
10. Escalate to senior management and relevant legal/compliance teams if required.

P2 – High-Severity Incidents

Assign experienced SOC analysts and appropriate technical specialists.

Examples:

- Administrator account compromise
- Active malware infection
- Exploited critical vulnerability

These incidents should receive rapid investigation and continuous monitoring.

P3 – Medium-Severity Incidents

Assign normal SOC analysts who can investigate and resolve the incident according to the defined SLA.

P4 – Low-Severity Incidents

These can be handled through standard SOC workflows, automation, or queue-based investigation.

However, low severity does not mean that the ticket should be ignored. It must still be tracked until closure.

G. Escalation of High-Priority Incidents

The P1 database compromise should immediately trigger the organization's escalation procedure.

Escalation may occur when:

- A critical asset is compromised.
- Sensitive data may have been exposed.
- Privileged credentials are compromised.
- The attacker remains active.
- The incident exceeds its SLA.
- The incident affects multiple systems.
- Business operations are disrupted.

The escalation chain can be:

SOC Analyst → SOC Lead → Incident Response Manager → CISO/Security Manager → Senior Management

Legal, compliance, privacy, communications, or external incident-response teams may also be involved depending on the incident.

The ticket should record **who was contacted, when they were contacted, why escalation occurred, and what decisions were made.**

H. SLA Management

Every incident should have an SLA based on its severity.

For example:

Priority	Example Response Target	Monitoring
P1	Immediate	Continuous
P2	Within minutes	Frequent
P3	Within defined business SLA	Regular
P4	Within standard service window	Normal queue

The exact targets should be defined by organizational policy.

The ticketing system should generate automatic alerts when:

- An SLA deadline is approaching.
- An SLA has been breached.
- An incident has remained inactive for too long.
- A critical incident has not been updated.
- Required escalation has not occurred.

I. SOC Coordination and Resolution

The incident response manager should coordinate all relevant teams.

For a database compromise, for example:

SOC

→ Detects and investigates the alert

Network Team

→ Blocks malicious connections

Database Team

→ Secures and investigates the database

IAM Team

→ Disables compromised accounts

Threat Intelligence Team

→ Analyzes attacker indicators

Management

→ Makes business and risk decisions

All actions should be documented in the incident ticket.

Before closing the ticket, the SOC should verify:

- Threat has been removed.
- Vulnerability has been fixed.
- Compromised accounts have been secured.
- Systems have been restored.
- Monitoring shows no continuing malicious activity.
- Required evidence has been preserved.
- Business owner confirms recovery.

The incident can then be marked **Resolved** and eventually **Closed** after the required monitoring period.

2. Incident Ticket Review and Continuous Improvement

Introduction

If management observes repeated delays and SLA violations, historical incident tickets should be analyzed to determine why incidents are taking too long to resolve. The objective is not simply to identify which analysts were slow, but to identify **process, technology, staffing, communication, escalation, and workflow problems**.

A. Historical Ticket Review

The first step is to collect historical ticket data for a suitable period, such as the previous six months.

The following information should be extracted:

- Incident ID
- Incident type
- Severity
- Priority
- Detection time
- Ticket creation time
- Assignment time
- First-response time
- Escalation time
- Containment time
- Resolution time
- Closure time
- Assigned team
- SLA target
- SLA breach status
- Number of reassignment events
- Number of escalations
- Root cause
- Resolution method
- Recurring incident indicator

This creates a complete historical dataset for analysis.

B. Analyze Response and Resolution Times

Important metrics include:

Mean Time to Detect (MTTD)

Measures how long it takes to detect an incident.

$$\text{MTTD} = \text{Total Detection Time} / \text{Number of Incidents}$$

Mean Time to Respond (MTTR)

Measures how quickly the security team begins responding after detection.

$$\text{MTTR} = \text{Total Response Time} / \text{Number of Incidents}$$

Mean Time to Contain (MTTC)

Measures the time required to stop the incident from spreading or causing additional damage.

Mean Time to Resolve

Measures the average time from incident creation/detection until resolution.

SLA Breach Rate

$$\text{SLA Breach Rate} = (\text{Number of SLA-breached incidents} / \text{Total incidents}) \times 100$$

For example, if 20 of 200 incidents breached their SLA:

$$\text{SLA Breach Rate} = (20/200) \times 100 = 10\%$$

C. Identify Recurring Problems

Historical tickets should be grouped by common causes.

For example:

Problem	Evidence in Tickets	Possible Cause
Delayed assignment	Long time between creation and assignment	Insufficient SOC staffing
Repeated reassignment	Tickets moved between teams	Unclear ownership
SLA violations	High breach percentage	Unrealistic SLA or poor workflow
Delayed escalation	High escalation time	Unclear escalation criteria

Problem	Evidence in Tickets	Possible Cause
Repeated incidents	Same vulnerability appears repeatedly	Root cause not permanently fixed
Missing evidence	Incomplete ticket records	Poor documentation practices
Long investigation time	Similar alerts take hours	Lack of automation or skilled staff

This allows management to distinguish individual incidents from **systemic problems**.

D. Analyze Escalation Procedures

The organization should review:

1. Was the correct severity assigned?
2. Was escalation triggered at the correct time?
3. Were the correct teams notified?
4. Was management informed when required?
5. Did analysts know the escalation criteria?
6. Were escalation contacts available?
7. Did escalation actually reduce resolution time?

If several critical incidents experienced delayed escalation, the escalation matrix should be redesigned.

For example:

P1 → Immediate SOC Lead + Incident Manager + CISO notification

P2 → SOC Lead + Relevant Technical Team

P3 → Standard SOC workflow

P4 → Normal queue/automated handling

E. Identify Bottlenecks

The ticket lifecycle should be examined stage by stage:

Detection → Ticket Creation → Assignment → Investigation → Escalation → Containment → Eradication → Recovery → Closure

Suppose the analysis shows:

- Ticket creation: 5 minutes
- Assignment: 45 minutes
- Investigation: 60 minutes
- Containment: 20 minutes
- Recovery: 30 minutes

The biggest bottleneck is clearly **assignment**.

Therefore, adding more investigation tools alone would not solve the problem. The organization should improve the assignment and routing process.

F. Recommended Improvements

1. Automated Ticket Assignment

Use ticket-management automation to route incidents based on:

- Severity
- Incident category
- Affected technology
- Analyst specialization
- Current workload

This reduces manual delays.

2. Automated SLA Monitoring

Configure automatic notifications for:

- New high-priority incidents
- Approaching SLA deadlines
- SLA breaches
- Unassigned tickets
- Tickets without recent updates

Critical tickets should receive stronger alerts and management visibility.

3. Standardized Incident Playbooks

Create playbooks for common incidents such as:

- Phishing
- Malware
- Credential compromise
- Ransomware
- Data breach
- Unauthorized access
- Vulnerability exploitation

A playbook provides analysts with predefined investigation, containment, escalation, and recovery steps.

4. Clear Ownership

Every incident must have a clearly identified owner.

The ticket should show:

Incident Owner → Responsible Team → Escalation Manager → Business Owner

This prevents incidents from being repeatedly transferred between teams.

5. Improve SOC Staffing and Training

If historical data shows that incidents remain unassigned or unresolved during peak periods, management should evaluate:

- SOC staffing levels
- Shift coverage
- Analyst workload
- Required technical skills
- Training requirements

Training should focus on incident types that repeatedly cause delays.

6. Introduce Automation

Security automation can reduce response time.

Examples include:

- Automatically disabling compromised accounts
- Automatically isolating infected endpoints

- Blocking malicious IP addresses
- Enriching IOCs with threat intelligence
- Automatically creating tickets from SIEM alerts
- Automatically assigning tickets based on severity

Automation should be carefully controlled so that high-impact actions have appropriate safeguards.

7. Improve Ticket Documentation

Mandatory ticket fields should include:

- Incident description
- Severity
- Affected asset
- Evidence
- IOCs
- Investigation steps
- Actions performed
- Escalation history
- SLA status
- Resolution
- Root cause
- Lessons learned

Required fields prevent incomplete records.

G. Continuous Improvement Cycle

Historical ticket analysis should become a continuous process:

Collect Ticket Data

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Measure KPIs

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Identify Bottlenecks

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Perform Root Cause Analysis

↓

Implement Corrective Actions

↓

Monitor Results

↓

Review Again

This ensures that the incident-response process continuously improves rather than repeating the same mistakes.

H. Example Improvement Dashboard

Metric	Previous	Target	Action
Average assignment time	45 min	<10 min	Automate ticket routing
Average response time	60 min	<30 min	Improve SOC workflow
Average resolution time	8 hrs	<4 hrs	Introduce playbooks/automation
SLA breach rate	15%	<5%	Automated SLA alerts
Reassignment rate	20%	<10%	Define ownership rules
Recurring incidents	12%	<5%	Strengthen root-cause remediation

Management should review these metrics periodically to determine whether the implemented improvements are actually reducing delays.

Conclusion

Effective incident management requires more than simply creating tickets. Each incident must be **accurately classified, documented, prioritized, assigned, tracked, escalated, and resolved according to its severity and SLA**. When multiple incidents occur simultaneously, resources should be concentrated on critical incidents while lower-priority incidents continue through controlled queues and automated workflows.

For repeated SLA violations, historical ticket analysis should identify delays in **assignment, investigation, escalation, containment, and resolution**. Metrics such as MTTD, MTTR, Mean Time to Contain, resolution time, SLA breach rate, and reassignment rate help identify bottlenecks.

The organization can then improve its ticket-management process through **automated routing, SLA alerts, standardized playbooks, clear ownership, better staffing and training, automation, and improved documentation**.

This approach creates a measurable continuous-improvement cycle and ensures that future incidents are handled faster, more consistently, and with better coordination between the SOC and other technical and management teams.